WIFFLE BALL

Pitching Physics for Newer Players

Illustrated guide to holes, finger placement, and throws — built for brand-new unscuffed balls and a 55 mph speed limit.

Big idea: You do not need to throw harder to make the ball move. With a new official Wiffle ball, movement comes from hole direction, a clean grip on the seam, and a repeatable arm slot.

What you will learn

- How the 8 holes create curve (outside turbulence + inside vortices)
- Where the holes should face for common pitches
- Where your fingers go (and what not to cover)
- How to throw each pitch at ≤ 55 mph
- Why “no scuff marks” means you must be sharper with orientation

1. Ball Anatomy: Where the Holes Go

An official Wiffle ball is not symmetric. One hemisphere has **eight oblong holes**. The other half is smooth plastic. That uneven shape is the whole pitching game. Air treats each half differently, so the ball refuses to fly straight unless you ask it to.

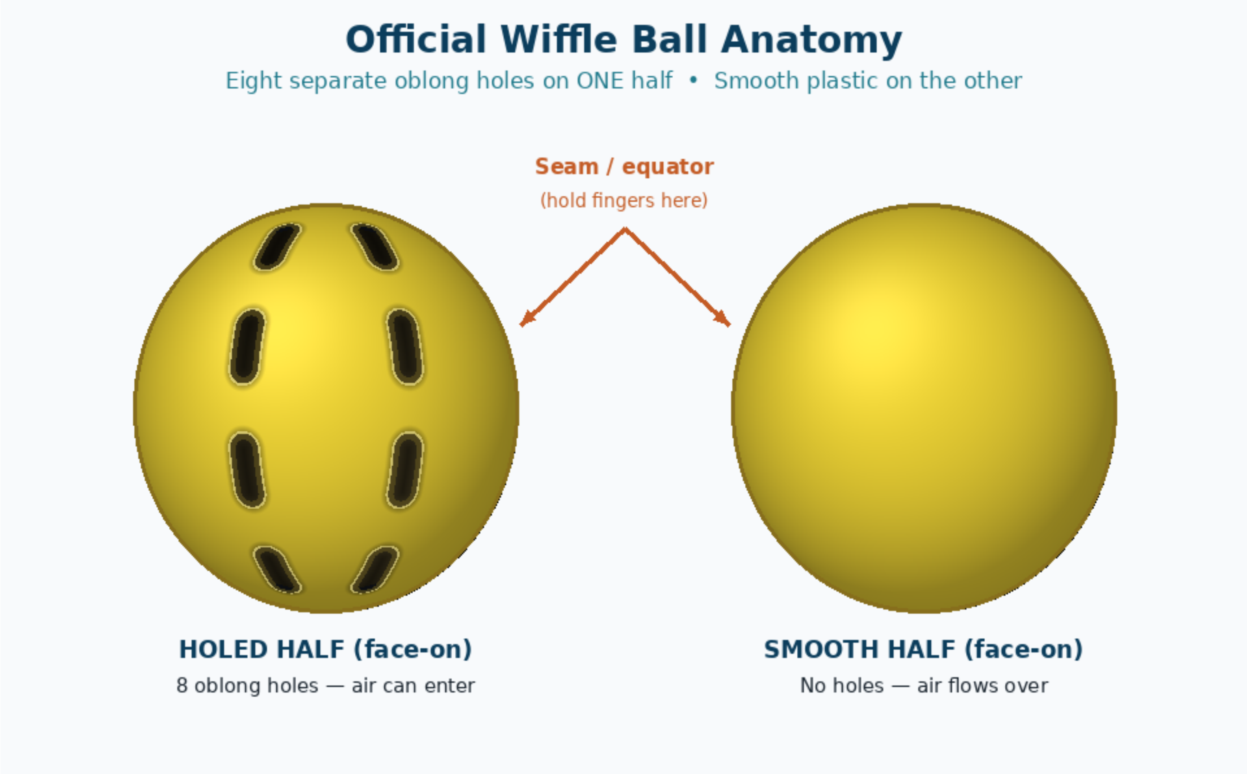

Figure 1. Holed half vs smooth half. Hold near the seam; keep holes open.

- **Holed half:** air can rush inside and create swirling pockets (vortices).
- **Smooth half:** air flows over the plastic more cleanly.
- **Seam / equator:** the dividing line — this is your finger home base.

2. The Physics (Without the Heavy Math)

Researchers (including wind-tunnel work popularized from Lafayette College studies) describe two cooperating effects:

- **Outside the ball:** holes stir the air, creating extra turbulence and uneven drag/lift on one side.
- **Inside the ball:** air entering the holes can form **trapped vortices** that push from within.
- **Net force:** those inside and outside forces constantly shift. Their sum is the break you see.

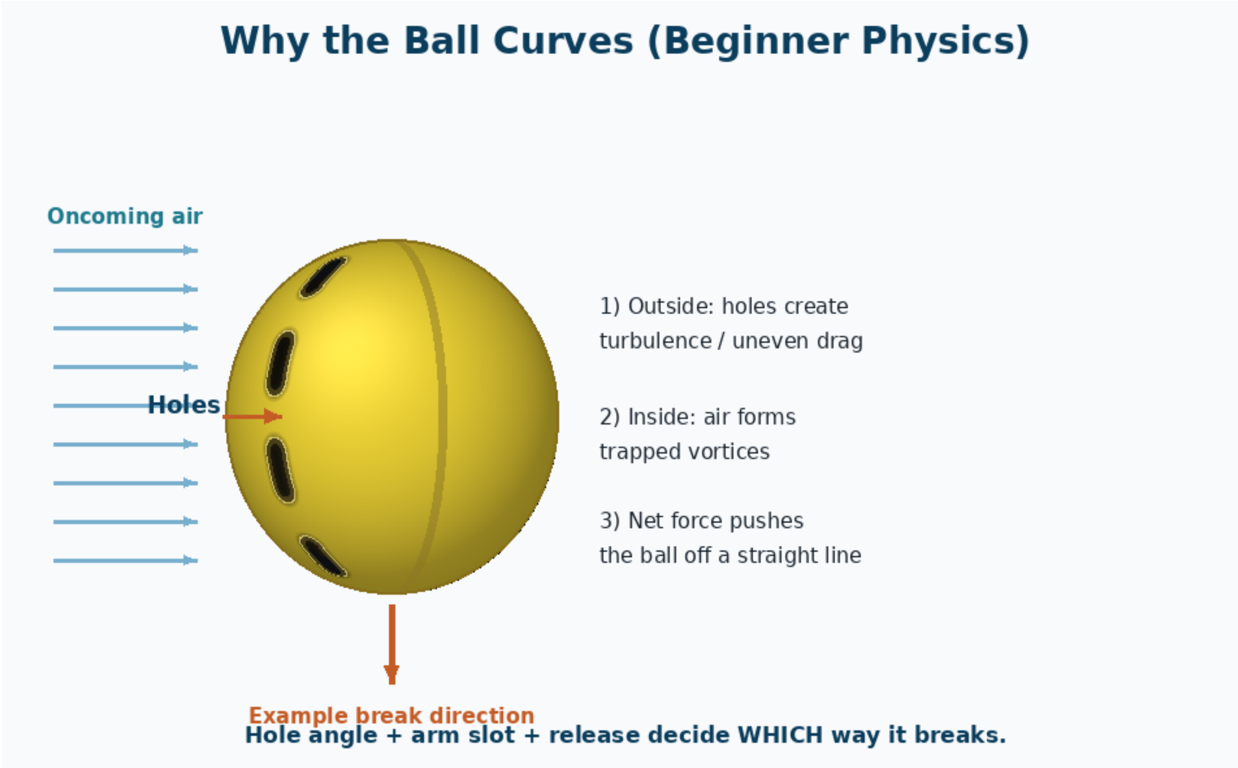

Figure 2. Asymmetric airflow is why orientation matters more than arm strength.

> *At 55 mph and under, you are in a sweet learning zone: hard enough for real movement, soft enough that overthrowing usually makes command worse, not break better.*

3. Hole Orientation Map

Before every pitch, ask: **Which way are the holes facing?** Classic package tips put holes up for straighter, toward the thumb for curve, and toward the outer fingers for slider. Modern backyard systems use the same idea with left/right + arm slot.

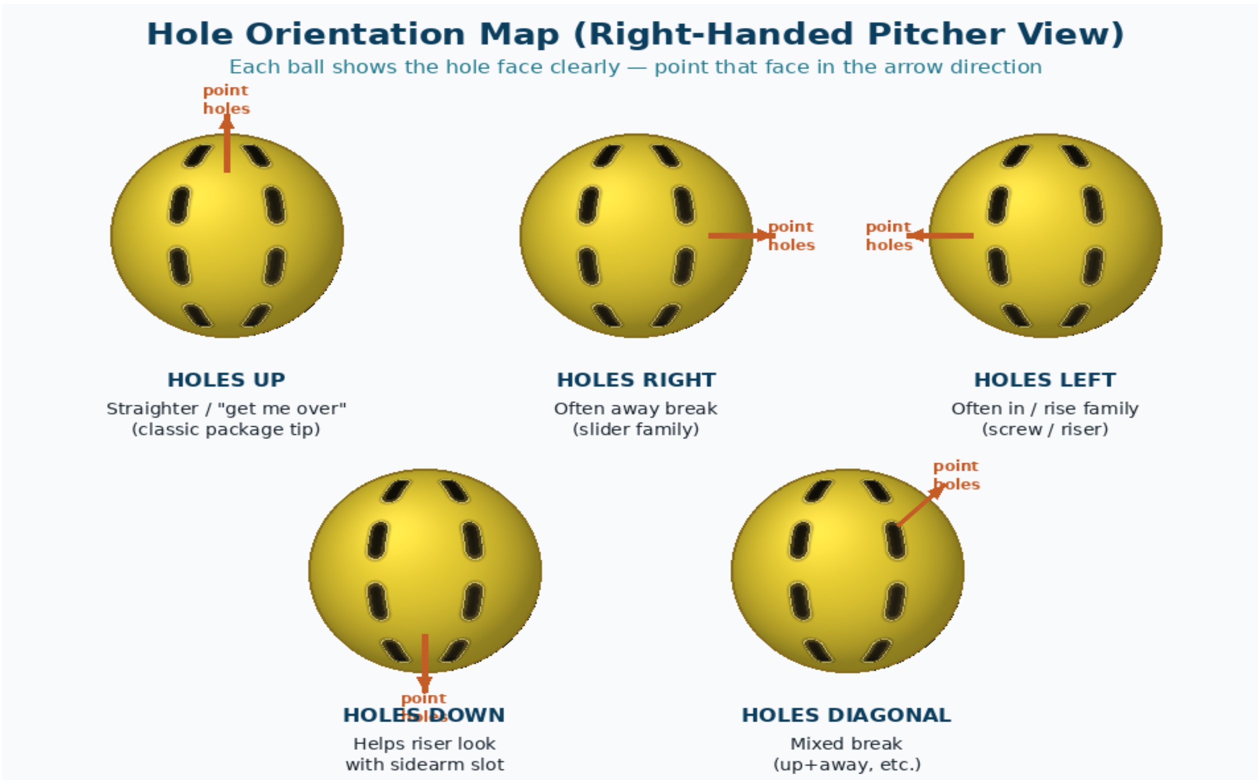

Figure 3. Learn these five looks first. Same arm speed, different holes → different pitch.

4. Finger Placement: The Basic Grip

Most beginner pitches use the same hand shape. Change the **hole direction** and **arm slot**, not a brand-new claw every time.

- Index and middle fingers rest on the seam (the equator).
- Thumb supports underneath on the smooth half.
- Hold loosely — death grip adds bad spin and blocks feel.
- **Never smother the holes** with your fingers. Open holes = more air in = more movement.

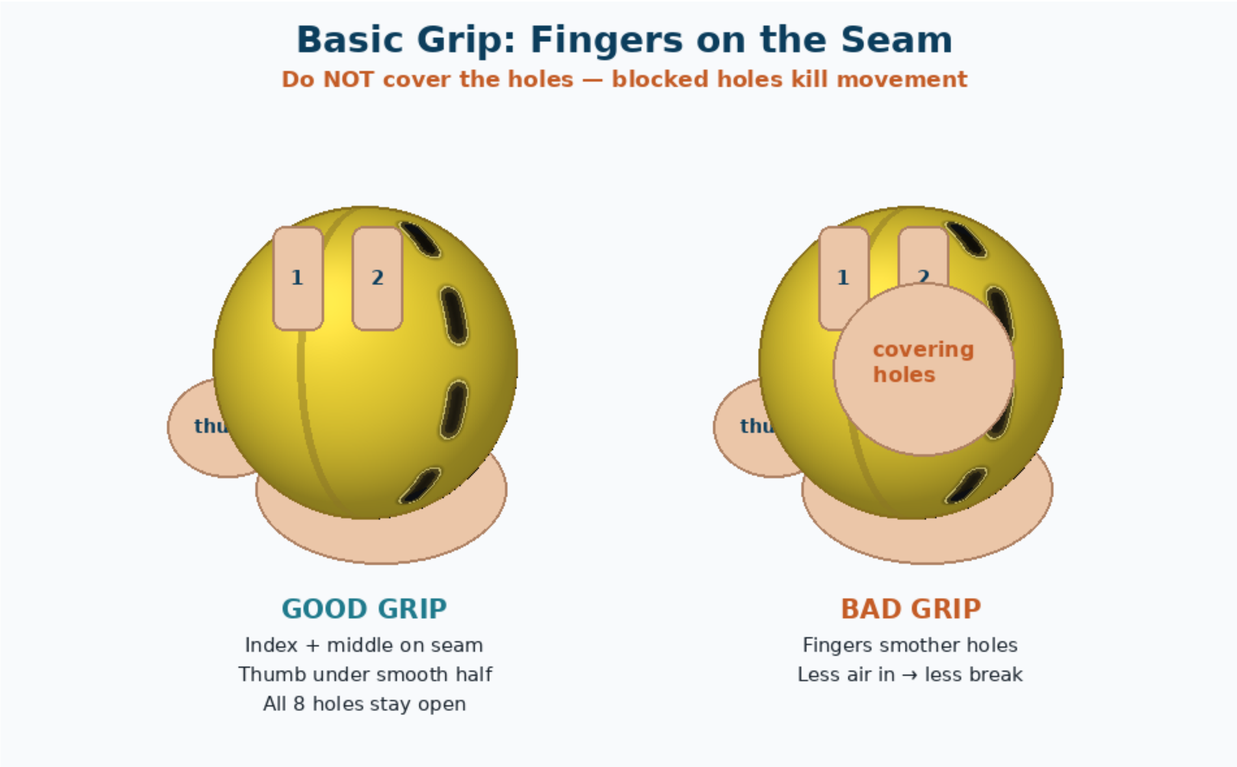

Figure 4. Good seam grip vs blocked-hole grip.

5. Starter Pitches: Holes + Fingers + Throw

All of these assume a **brand-new ball** (no scuffs) and staying at or under **55 mph**. Directions below are for a **right-handed pitcher**. Lefties mirror left/right.

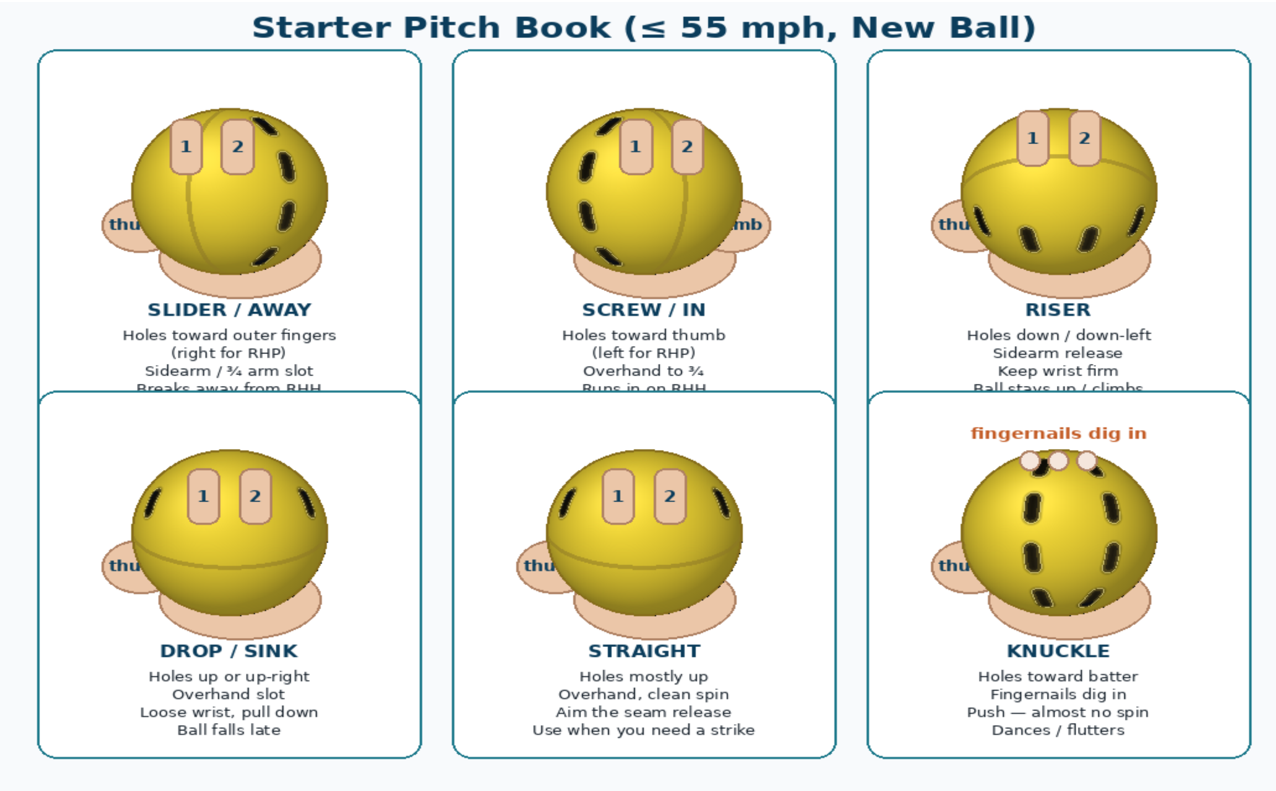

Figure 5. Six beginner pitches with hole direction and finger cues.

Pitch	Holes	Fingers / throw	What to expect
Straight	Mostly up	Seam grip, overhand, clean release	Strike-steal pitch
Slider / away	Toward outer fingers (R)	¾ or sidearm; stay online	Breaks away from RHH
Screw / in	Toward thumb (L)	Overhand/¾; firm wrist	Runs in on RHH
Riser	Down or down-left	Sidearm; throw through high target	Stays up / late rise look
Drop	Up or up-right	Overhand; soft pull-down finish	Late fall
Knuckle	Toward batter	Nails dig; push with almost no spin	Flutter / dance

6. How to Throw It (Arm Slot Pictures)

Think of a tiny checklist on the mound: **holes** → **slot** → **target**. Do not try to “help” the break by yanking your wrist sideways every pitch. A clean release with the holes set correctly does most of the work — especially under 55 mph.

Figure 6. Overhand, three-quarter, and sidearm — match the slot to the pitch family.

Release cues that help new players

- Pick a glove-sized target and throw through it (not “at the break”).
- Keep the same arm speed between pitches; change holes/slot instead of muscling up.
- Finish your throw — stopping your hand early often kills both aim and movement.
- If the pitch spins like a helicopter, your fingers slid across the ball. Reset to seam grip.

7. New Balls + 55 mph: What That Means

Some backyard pitchers scuff or scratch balls to sharpen break. **You cannot do that here.** Only brand-new Wiffle balls with no scuff marks are legal, and velocity caps at 55 mph. Good news: that is still plenty for real movement if your orientation is honest.

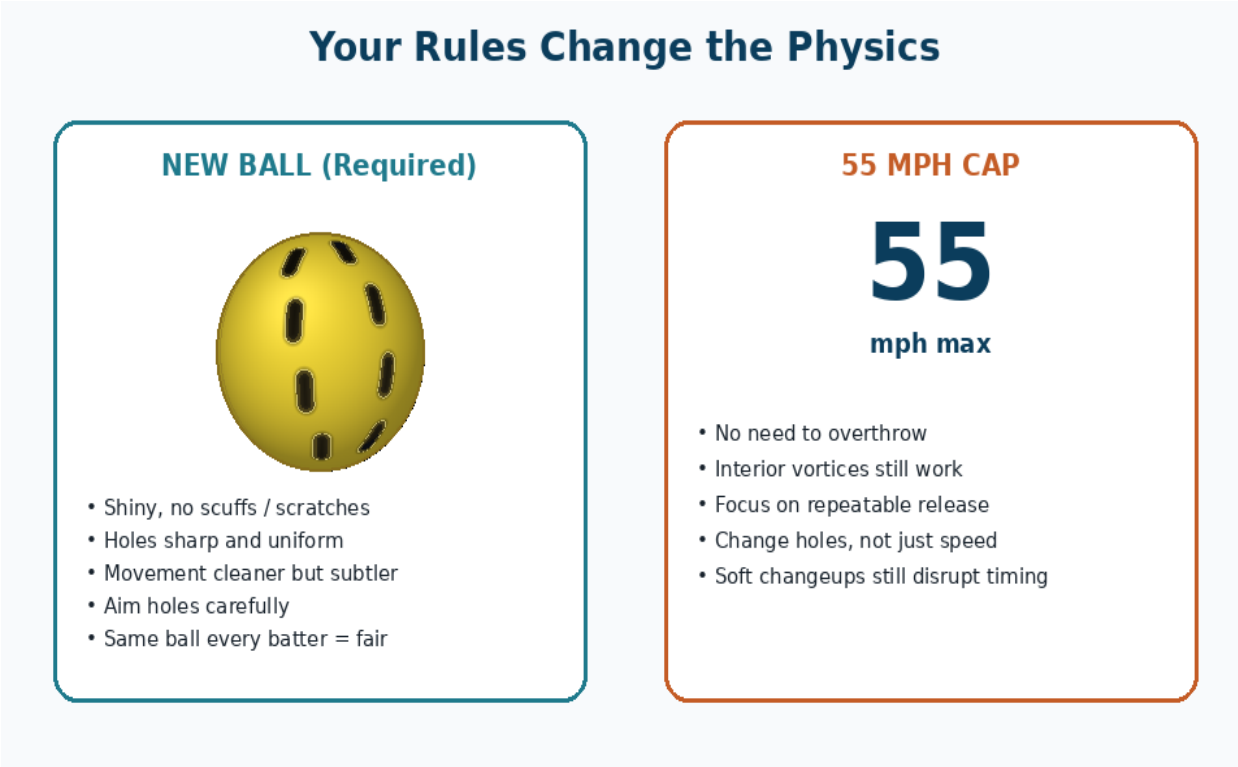

Figure 7. Your constraints are also your training plan: precision over doctoring.

- **No scuffs** means less “free” turbulence — so hole aim and seam grip matter more.
- **55 mph max** means command and sequencing beat raw power.
- Practice at game speed. If you only warm up at 30 and then jump to 55, orientation drifts.
- Rotate new balls as required; do not “wear one in.” Learn to win with fresh plastic.

8. 15-Minute Practice Plan

- **Minutes 1–3:** Seam grip only. 10 throws holes-up for strikes.
- **Minutes 4–7:** Holes-right slider family. Same arm speed, sidearm/¾.
- **Minutes 8–11:** Holes-left screw/riser family. Notice the different flight.
- **Minutes 12–14:** Alternate straight / away / in. Call the pitch before you throw.
- **Minute 15:** 5 knuckle pushes for feel (accept some wildness).

> *Troubleshooting: If nothing moves, check three things in order — (1) are holes open? (2) are holes actually facing the direction you think? (3) are you throwing the slot that matches the pitch?*

9. Quick Pocket Card

- New ball, no scuffs, ≤ 55 mph.
- Fingers on the seam. Holes uncovered.
- Holes up → straighter. Holes outer → away. Holes thumb → in/curve.
- Sidearm helps slider/riser looks. Overhand helps drop/straight.
- Throw through the target. Let the ball's shape create the break.

Master orientation first. Speed is capped — craft is not.